



**MAT2072 – Numerical Analysis I**  
**Chapter 2: Solutions of Equations in One Variable**  
**SEMESTER I – 2024/2025**

Department of Mathematics, Faculty of Science  
University of Peradeniya

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## 1 Introduction

Many problems in science, engineering, economics, and applied mathematics can be reduced to finding solutions of an equation of the form

$$f(x) = 0, \tag{1}$$

where  $f$  is a given function and the value  $x$  satisfying the equation is called a **root** or **zero** of  $f$ .

In many practical situations, the roots of an equation cannot be obtained analytically. For example, equations such as

$$x + e^{-x} + 2 = 0, \tag{2}$$

$$x^3 + x - 1 = 0, \tag{3}$$

$$2 \cos x + 3x^2 + 4^x = 0. \tag{4}$$

do not possess simple closed-form solutions. In such cases, numerical methods are employed to obtain approximate solutions.

The process of determining approximate roots of equations is known as **root-finding**.

## 2 Mathematical Preliminaries

### 2.1 Root of an Equation

**Definition.** A number  $r$  is called a **root** (or **zero**) of the equation

$$f(x) = 0 \tag{5}$$

if

$$f(r) = 0. \tag{6}$$

### 2.2 Example

Consider

$$f(x) = x^2 - 4. \tag{7}$$

The roots are

$$x = 2 \quad \text{and} \quad x = -2, \tag{8}$$

since

$$f(2) = 0 \quad \text{and} \quad f(-2) = 0.$$

### 3 Numerical Methods for Solving Nonlinear Equations

Numerical methods for solving equations can generally be classified into two categories.

#### 3.1 Bracketing Methods

These methods begin with an interval containing a root and repeatedly reduce the interval length.

Examples include:

- Bisection Method
- False Position Method

#### 3.2 Open Methods

These methods start from one or more initial approximations and generate successive approximations to the root.

Examples include:

- Fixed-Point Iteration Method
- Newton-Raphson Method
- Secant Method

## 4 Existence of Roots

Before applying a numerical method, it is important to determine whether a root exists in a given interval.

The following theorem provides a useful criterion.

### 4.1 Intermediate Value Theorem

If  $f(x)$  is continuous on  $[a, b]$  and if  $f(a)$  and  $f(b)$  are of opposite signs, then there exists at least one number  $P \in (a, b)$  such that

$$f(P) = 0.$$

## 4.2 Interpretation

If a continuous function changes sign over an interval, then it must cross the  $x$ -axis somewhere within that interval.

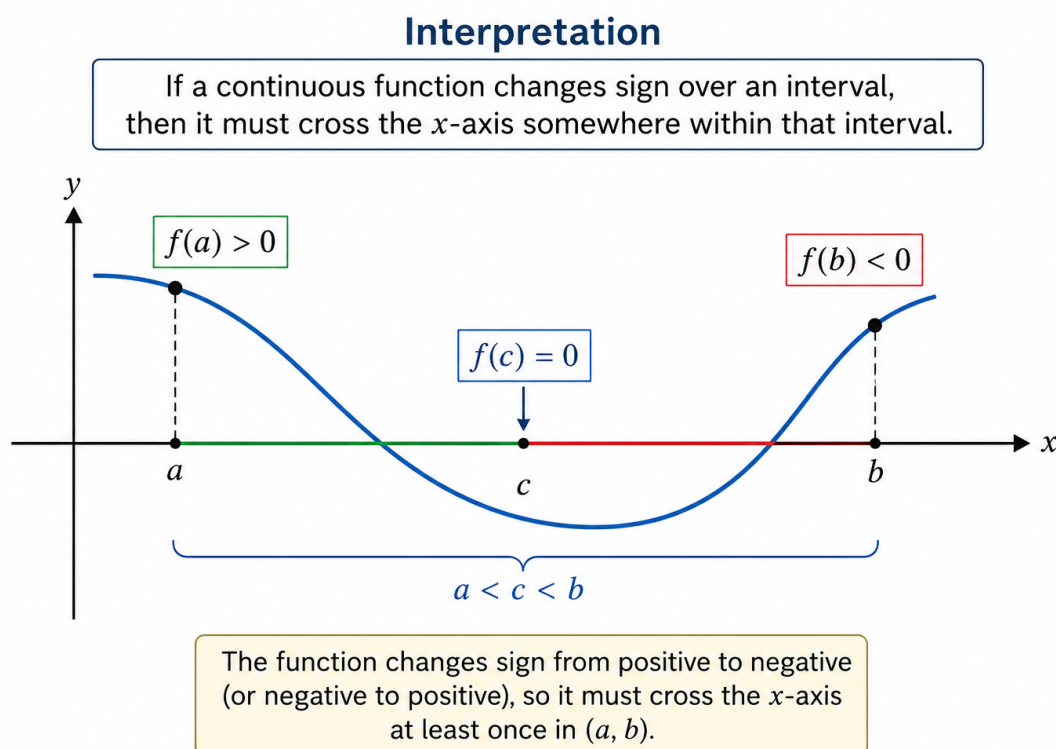


Figure 1: Intermediate Value Theorem

## 4.3 Example

Consider

$$f(x) = x^3 - x - 2. \quad (9)$$

Evaluating at the endpoints,

$$f(1) = 1 - 1 - 2 = -2,$$

and

$$f(2) = 8 - 2 - 2 = 4.$$

Since  $f(1)$  and  $f(2)$  are of opposite signs, then there exists at least one number  $P \in (1, 2)$  such that

$$f(P) = 0.$$